# WETLAND DEGRADATION ASSESSMENT

Assessment of ecological change within the selected analysis area

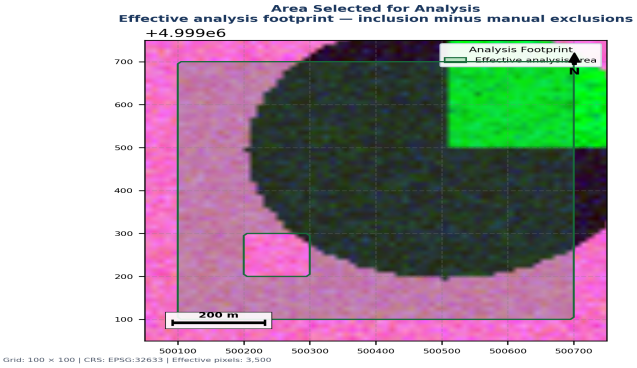
| OBSERVATION PERIOD | 2023-08-14 → 2023-08-28 | INTERVAL | 14 days |
|---|---|---|---|
| WETLAND CONTEXT | Project-provided wetland boundary | RUN ID | run_20260922_082400 |
| ANALYSIS FOOTPRINT | 100.00% active area (35.00 ha) | SPATIAL REFERENCE | EPSG:32633 |

OVERALL WETLAND ASSESSMENT

## NO MEASURABLE WETLAND DEGRADATION DETECTED

Available multi-stream remote-sensing evidence demonstrates no systemic wetland degradation within the selected analysis area across the observation window.

| WETLAND-LEVEL INDICATOR | OBSERVED RESULT |
|---|---|
| Wetland Degradation (Hydro-Ecological) | 0.00% |
| Wetland Recovery (Hydro-Ecological) | 0.00% |
| Stable Classified State | 100.00% |
| Vegetation Degradation (Supporting Evidence) | 0.00% |
| Vegetation Recovery (Supporting Evidence) | 0.00% |
| Surface-Water Loss (Contraction) | 0.00% |
| Continuous Spectral Shift (Mean CVA) | 0.6969 |
| Effective Analysis Area | 100.00% (35.00 ha) |




*Analysis Area Footprint (Restricted AOI)*

# 1. THE AREA WE ANALYZED

This assessment examines changes within the area selected for analysis. Vegetation condition, spectral indicators and surface-water evidence are compared between the two observation dates. The selected area defines where the assessment is performed. It should not automatically be interpreted as a legal or statutory wetland boundary.

The analysis is spatially bounded by the configured wetland boundary. All ecological condition indicators are evaluated within this defined area.

| SPATIAL ATTRIBUTE | OBSERVED / CONFIGURED VALUE |
|---|---|
| **Wetland context status** | WETLAND_BOUNDARY_AVAILABLE |
| **Wetland source** | Project-provided wetland boundary |
| **Source dataset / boundary** | wetland_boundary.geojson |
| **Selected analysis area** | 100.00% (36.00 ha) |
| **Effective analysis footprint** | 100.00% (35.00 ha) |
| **Manual exclusion footprint** | 2.78% (1.00 ha) |
| **Physical wetland loss pathway** | NOT_ASSESSED |

### SOURCE IMAGERY (FALSE-COLOUR CIR COMPOSITES)

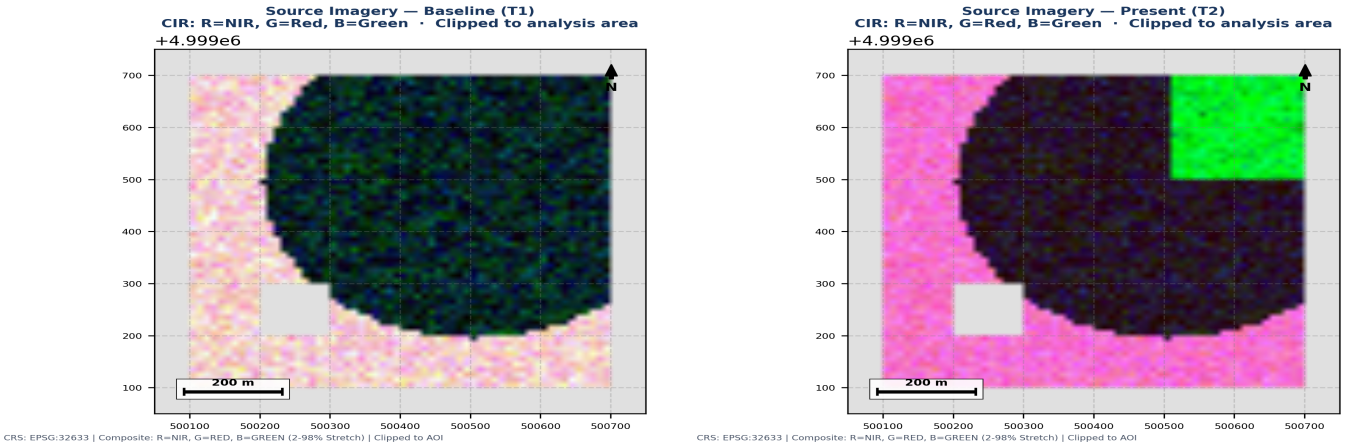



*Figure 1: Baseline (T1, 2023-08-14) and present (T2, 2023-08-28) multispectral false-colour composites (NIR-Red-Green CIR), restricted to the effective analysis area.*

### ANALYSIS FOOTPRINT AUDIT

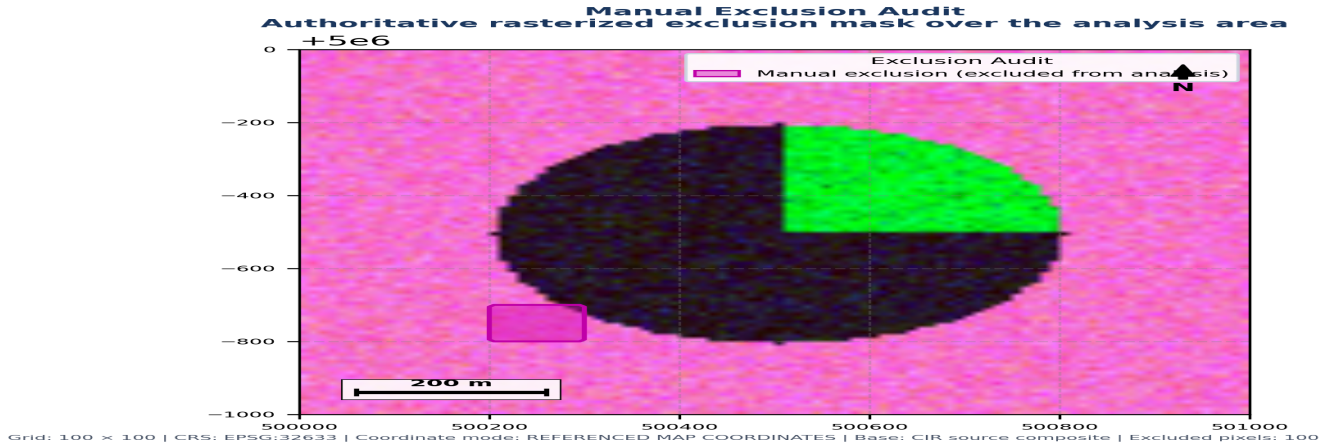



*Figure 2: Authoritative exclusion audit map showing verified exclusions (2.78% of footprint removed).*

# 2. WETLAND CONDITION

Wetland condition is evaluated by integrating multiple independent evidence streams. Vegetation condition, continuous spectral dynamics, and surface-water presence are each assessed across the active analysis footprint. The table below outlines how each evidence stream informs the integrated wetland assessment.

| EVIDENCE STREAM | OBSERVED FINDING | ROLE IN ASSESSMENT |
|---|---|---|
| **Vegetation condition** | Stable canopy state (100.00% persistent) | Tracks canopy structure and foliar condition class shifts. |
| **Spectral change** | Mean CVA magnitude: 0.6969 (ΔNDVI: +0.4944) | Measures continuous, sub-classification physiological canopy displacement. |
| **Surface-water change** | Loss: 0.00%; Gain: 10.80% | Independently evaluates wetting and drying trends. |
| **Integrated assessment** | **NO MEASURABLE WETLAND DEGRADATION DETECTED** | Synthesizes multi-stream evidence into the wetland degradation finding. |

Evaluating wetland condition requires distinguishing transient canopy fluctuations from systemic hydro-ecological loss. Optical vegetation stress without concurrent hydrological contraction indicates vegetation-level dynamics, whereas concurrent vegetation stress and water loss provide evidence of wetland degradation.

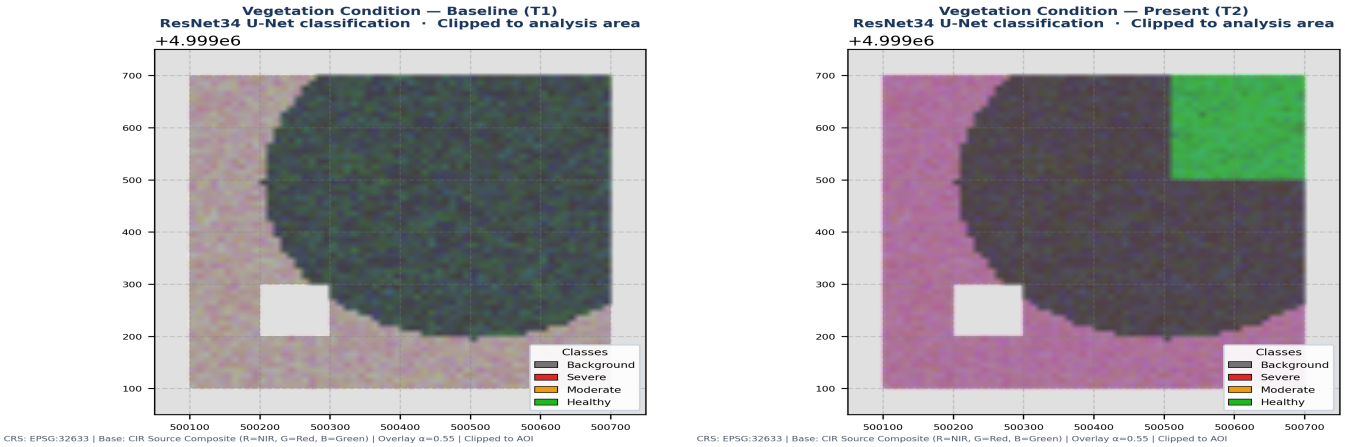



*Figure 3: Classified vegetation condition at baseline (T1, 2023-08-14, left) and present (T2, 2023-08-28, right), restricted to the effective analysis footprint.*

# 3. VEGETATION EVIDENCE

Vegetation condition serves as an essential line of evidence in the wetland assessment. Canopy condition is classified at baseline (T1) and present (T2) into four discrete states: Healthy Vegetation, Moderate Stress, Severe Stress, and Background / Non-Vegetated. All proportions are reported as percentages of the effective analysis footprint.

| CONDITION CLASS | BASELINE (T1) % | PRESENT (T2) % | NET CHANGE (%) |
|---|---|---|---|
| **Healthy vegetation** | 0.00% | 0.00% | 0.00% |
| **Moderate stress** | 0.00% | 0.00% | 0.00% |
| **Severe stress** | 0.00% | 0.00% | 0.00% |
| **Background / non-vegetated** | 285.71% | 285.71% | 0.00% |

### VEGETATION TRANSITION DYNAMICS

Transitions between observation dates trace the exact trajectory of vegetation shifts, separating movement toward higher stress (degradation pathways) from recovery or canopy loss.

| TRANSITION PATHWAY | DEFINED TRANSITIONS | FOOTPRINT PROPORTION |
|---|---|---|
| **Persistent classified state** | H→H, M→M, S→S, B→B | 100.00% |
| **Movement toward poorer vegetation condition** | H→M, H→S, M→S | 0.00% |
| **Movement toward better vegetation condition** | M→H, S→M, S→H | 0.00% |
| **Canopy loss / non-vegetated transition** | H→B, M→B, S→B | 0.00% |
| **Canopy emergence / colonisation** | B→H, B→M, B→S | 0.00% |

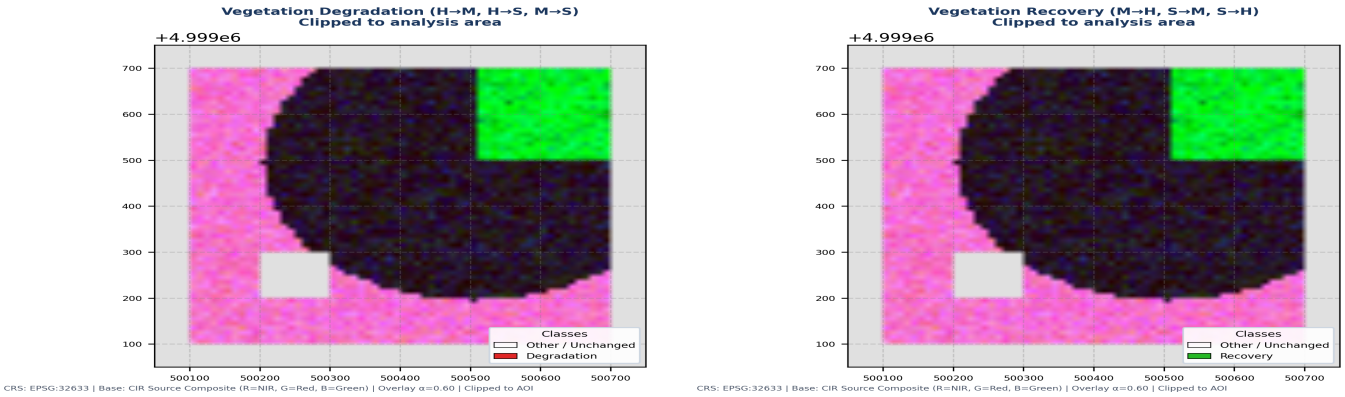



*Figure 4: Spatial distribution of detected vegetation degradation (left) and recovery (right), restricted to the effective analysis footprint.*

**Vegetation evidence summary:** 100.00% of the active analysis area remained in a stable classified condition state. 0.00% followed vegetation degradation pathways, and 0.00% followed recovery pathways.

# 4. SPECTRAL EVIDENCE

Continuous spectral indices provide independent evidence of foliar and physiological condition shifts. Change Vector Analysis (CVA) integrates ΔNDVI, ΔGNDVI, and ΔNDRE in a multi-dimensional feature space to capture subtle canopy adjustments that occur within or across categorical classification boundaries. These metrics represent continuous radiometric evidence rather than standalone wetland degradation measures.

| SPECTRAL INDICATOR | MEAN | MEDIAN | P95 |
|---|---|---|---|
| **CVA Magnitude (‖ΔV‖)** | 0.6969 | 0.6719 | 1.3386 |
| **ΔNDVI (ΔNDVI)** | 0.4944 | 0.2951 | 1.0850 |
| **ΔGNDVI (ΔGNDVI)** | 0.2689 | 0.0051 | 0.8085 |
| **ΔNDRE (ΔNDRE)** | -0.0334 | -0.0046 | 0.3494 |

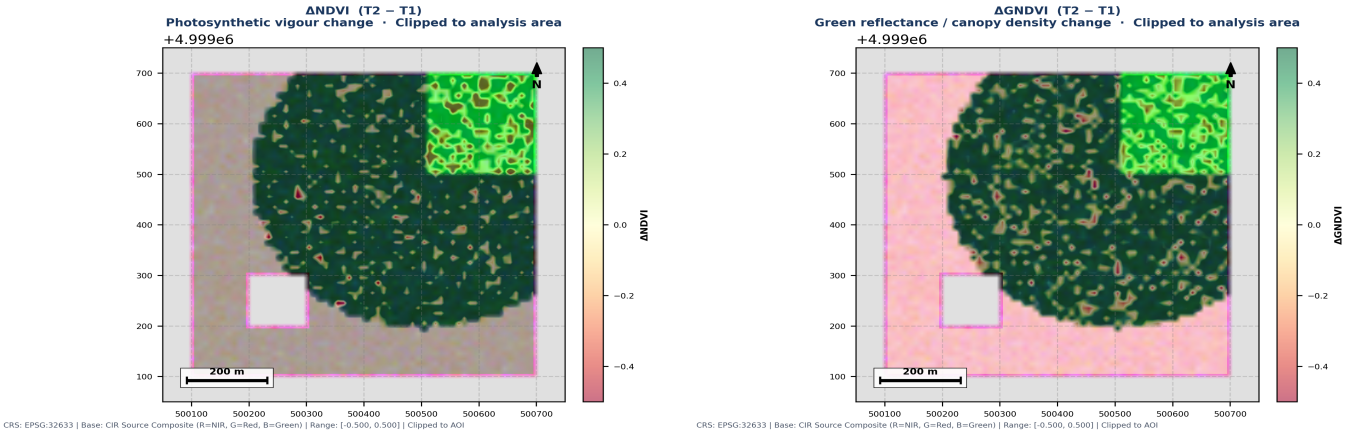



*Figure 5: Continuous vegetation index difference maps (ΔNDVI, left; ΔGNDVI, right) across the analysis footprint.*

**Spectral evidence summary:** Continuous radiometric monitoring records an average CVA displacement of 0.6969, reflecting sub-classification physiological variation across the vegetative canopy.

# 5. SURFACE-WATER EVIDENCE

Surface-water presence and moisture dynamics provide an essential hydrological line of evidence. Water presence is evaluated from the Normalized Difference Water Index (NDWI) using a calibrated threshold of 0.00. Where quantitative water extent cannot be derived from available observations, metrics are reported as 'Not available' to prevent misleading zero-loss assumptions.

| HYDROLOGICAL METRIC | FOOTPRINT PROPORTION | PHYSICAL EXTENT | DEFINITION |
|---|---|---|---|
| **Baseline Water Extent (T1)** | 0.77% | 0.27 ha | Surface water detected at baseline observation |
| **Present Water Extent (T2)** | 11.57% | 4.05 ha | Surface water detected at present observation |
| **Surface Water Expansion (Gain)** | 10.80% | 3.78 ha | Newly emerged surface water / wetting |
| **Surface Water Contraction (Loss)** | 0.00% | 0.00 ha | Previously inundated surface becoming dry |
| **NDWI Water Threshold** | 0.00 | N/A | Calibrated Green/NIR NDWI zero-plane threshold |

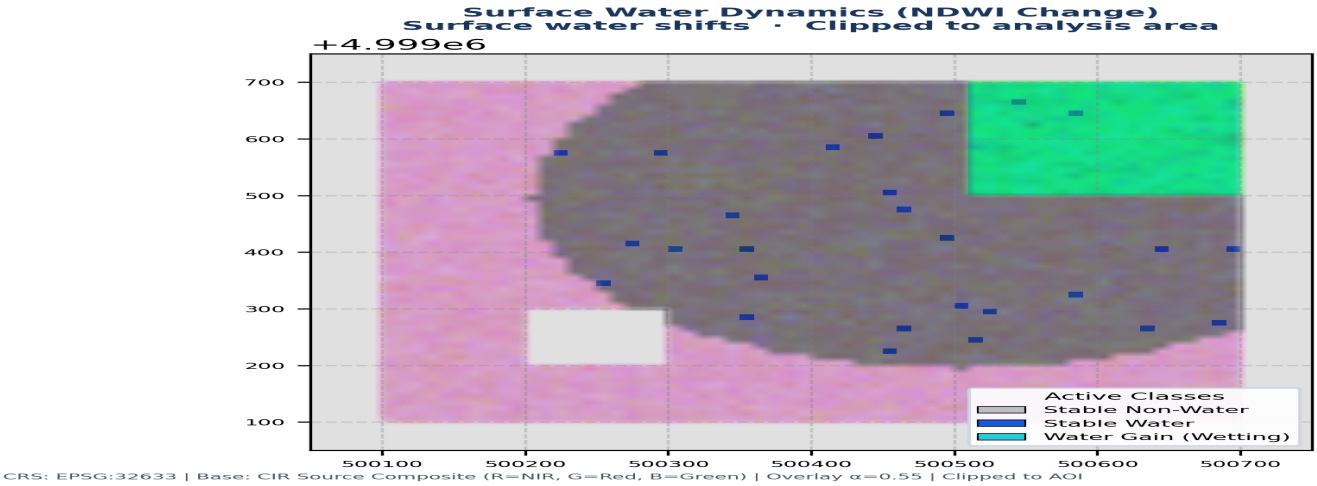



*Figure 6: Surface-water dynamics (wetting, drying, and stable water), restricted to the effective analysis footprint.*

**Hydrological evidence summary:** Surface-water dynamics indicate 0.00% contraction (drying) and 10.80% expansion (wetting) across the analyzed area.

# 6. WETLAND DEGRADATION ASSESSMENT

This section presents the primary analytical synthesis of the assessment. The evaluation synthesizes all available independent evidence streams—vegetation condition, continuous spectral shifts, and surface-water dynamics—to determine whether wetland degradation occurred within the selected analysis area between observation dates.

**Distinction between Vegetation Degradation and Wetland Degradation:**
Vegetation stress alone does not establish wetland degradation. A hydro-ecological wetland degradation signal is established only when concurrent evidence of both vegetation deterioration and surface-water loss is detected. Where surface-water dynamics are unavailable or stable, vegetation shifts represent vegetation-only condition responses (such as physiological stress or seasonal variation) rather than systemic wetland loss.

**PRIMARY ASSESSMENT: NO MEASURABLE WETLAND DEGRADATION DETECTED**

**Multi-evidence synthesis does not identify concurrent vegetation and hydrological decline sufficient to establish wetland degradation.**

**PHYSICAL WETLAND LOSS: NOT ESTABLISHED**

Physical wetland loss is not established. In accordance with strict scientific safeguards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

# 7. EVIDENCE FUSION

Evidence fusion cross-references vegetation transition pathways with surface-water dynamics. A composite hydro-ecological degradation signal requires concurrent evidence of both vegetation stress and surface-water loss, preventing single-sensor optical artifacts from triggering false alarms.

| EVIDENCE CLASS | FOOTPRINT PROPORTION | PHYSICAL AREA | DEFINITION & MEANING |
|---|---|---|---|
| **Stable landscape** | 100.00% | 35.00 ha | Persistent classified condition state across both observation dates |
| **Hydro-ecological degradation** | 0.00% | 0.00 ha | Concurrent vegetation degradation and surface-water loss |
| **Hydro-ecological recovery** | 0.00% | 0.00 ha | Concurrent vegetation recovery and surface-water gain |
| **Vegetation-only degradation** | 0.00% | 0.00 ha | Vegetation condition decline without concurrent hydrological loss |

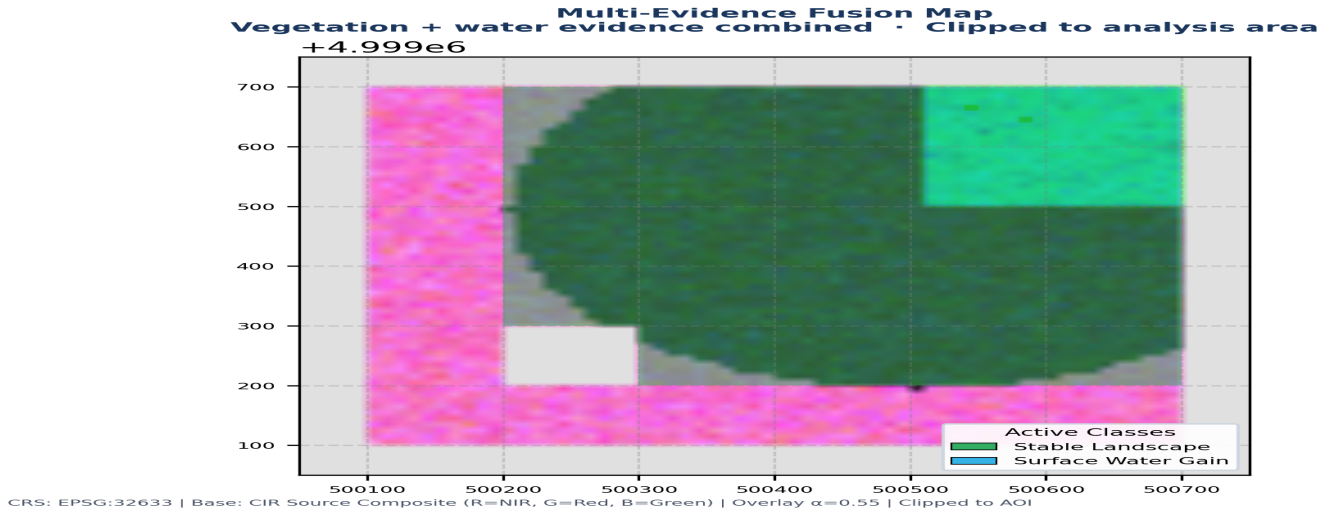



*Figure 7: Multi-evidence fusion combining vegetation transitions and surface-water dynamics, restricted to the effective analysis footprint.*

# 8. DATA QUALITY & SPATIAL ACCOUNTING

Spatial accounting verifies that every spatial component is traceable from source imagery through the inclusion boundary and manual exclusions to the effective analysis footprint.

| QUALITY CONTROL CHECK | STATUS | FINDING |
|---|---|---|
| **Spatial grid alignment** | PASS | T1 and T2 rasters verified on a common analysis grid without spatial distortion. |
| **Footprint reconciliation** | PASS | Selected analysis area fully reconciled with manual exclusions (100.00% accounted for). |
| **Independent ground reference** | UNAVAILABLE | No external ground-truth validation plots were supplied for this run. |
| **Physical depletion guard** | ENFORCED | Optical vegetation decline is prevented from being reported as physical wetland loss. |

### SPATIAL FOOTPRINT RECONCILIATION

| FOOTPRINT COMPONENT | PROPORTION OF SELECTED AREA | PHYSICAL AREA |
|---|---:|---:|
| **Original spatial overlap** | 100.00% | 100.00 ha |
| **Selected AOI / Inclusion** | 100.00% | 36.00 ha |
| **Manual exclusion** | 2.78% | 1.00 ha |
| **Effective analysis footprint** | 100.00% active | 35.00 ha |

# 9. TECHNICAL PROVENANCE & LIMITATIONS

Every metric reported in this assessment is derived directly from verifiable raster artifacts produced during the processing pipeline. No analytical values are interpolated or inferred without evidence.

| INDICATOR | OBSERVED VALUE | SPATIAL UNIT | PIPELINE ARTIFACT / METHOD |
|---|---|---|---|
| **Wetland source** | PROJECT_BOUNDARY | class | Wetland mask source selected by wetland context stage |
| **Wetland context status** | WETLAND_BOUNDARY_AVAILABLE | status | Wetland mask evaluation result |
| **Effective analysis area** | 100.00% | active footprint proportion | alignment/effective_analysis_mask.tif |
| **Manual excluded area** | 2.78% | footprint proportion | exclusions/manual_exclusion_mask.tif |
| **Vegetation degradation** | 0.00% | % of footprint | degradation/degradation_map.tif (H→M, H→S, M→S) |
| **Vegetation recovery** | 0.00% | % of footprint | degradation/recovery_map.tif (M→H, S→M, S→H) |
| **CVA Mean Magnitude** | 0.6969 | index units | change/cva_magnitude.tif (\|\|[ΔNDVI, ΔGNDVI, ΔNDRE]\|\|) |
| **Mean ΔNDVI** | 0.4944 | index units | change/delta_ndvi.tif (NDVI_T2 – NDVI_T1) |
| **NDWI Water Threshold** | 0.00 | index units | Calibrated Green/NIR NDWI threshold |
| **Physical depletion assessment** | NOT_ASSESSED | status | Independent-evidence claim guard |

### SCIENTIFIC LIMITATIONS & BOUNDARIES

**Observation Interval:** Remote-sensing comparisons reflect conditions at specific acquisition times. Short-term intervals capture immediate physiological or seasonal responses that may differ from multi-year ecological trends.
**Model Inferences:** Machine learning vegetation classifications are probabilistic estimates. In-situ field validation is recommended before using findings for regulatory, legal, or compensation purposes.
**Spatial Reference:** Where imagery lacks a validated Coordinate Reference System (CRS), proportional footprint metrics serve as the authoritative spatial unit. Physical area (hectares) is reported only when georeferencing is confirmed.

**Run:** run_20260922_082400 · **Software:** GIS Engine v1.0.0 · **Analysis Basis:** RASTER_CRS_AND_TRANSFORM